

MIXED-PRECISION NEURAL NETWORKS: A SURVEY

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ABSTRACT

Mixed-precision Deep Neural Networks achieve the energy efficiency and throughput needed for hardware deployment, particularly when the resources are limited, without sacrificing accuracy. However, the optimal per-layer bit precision that preserves accuracy is not easily found, especially with the abundance of models, datasets, and quantization techniques that creates an enormous search space. In order to tackle this difficulty, a body of literature has emerged recently, and several frameworks that achieved promising accuracy results have been proposed. In this paper, we start by summarizing the quantization techniques used generally in literature. Then, we present a thorough survey of the mixed-precision frameworks, categorized according to their optimization techniques such as reinforcement learning and quantization techniques like deterministic rounding. Furthermore, the advantages and shortcomings of each framework are discussed, where we present a juxtaposition. We finally give guidelines for future mixed-precision frameworks.

Keywords Deep Neural Networks · Mixed-Precision · Quantization · Computational Complexity

1 Introduction

With the thrive of Deep Neural Networks (DNNs) in the fields of object detection [1], autonomous cars [2], the internet of things (IoT) [3], genomics [4], and smart cities [5], questions started arising regarding the feasibility of deploying DNNs on hardware platforms limited by computational resources and/or power constraints. Even though they achieved high accuracy, DNNs have not been widely deployed yet on a large scale, especially on embedded systems and mobile phones. In hopes of mitigating the expensive overhead usually imposed by DNNs, many works in literature proposed "hardware-friendly" solutions. These include the following: pruning [6–11], new DNN designs [12–14], DNN architectures and hardware co-design [15–18], knowledge distillation [19–22], and quantization [23–33]. Quantization is the focus of this survey which is shown to be best way to achieve orders of improvement in the energy and a latency.

Quantization which dates back to 1990 [34] presents itself as a promising solution as it allows representing floating-point values of weights and/or activations and/or gradients in a fewer number of bits. As such, it reduces the memory footprint, computational complexity, and memory traffic volume [35–37], and renders itself suitable for support by existing hardware like CPUs and FPGAs.

Networks that have a 1-bit precision for weights and/or activations are known as Binary Neural Networks (BNNs) [38]. While Floating Point (FP) DNNs are the most accurate, BNNs are the most efficient. However, quantizing the DNN parameters into ultra-low precision severely degrades the accuracy. One way to mitigate this degradation is to carry out "fine-tuning", which is retraining for a short number of epochs, such as what was done in [28, 39, 40] and [29]. We denote quantization techniques that rely on fine-tuning for accuracy purposes as "retraining" quantization techniques. This introduces a new trade-off: accuracy vs. time consumption and computational complexity. In fact, there exists a factorial complexity in order to decide on the order of fine-tuning [32]. As such, works in literature like [41] and [42] have been proposed to quantize the network without fine-tuning, and these techniques became famous as "post-training" quantization techniques. But some of these techniques still suffer from an unexpected loss in performance. Moreover,

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